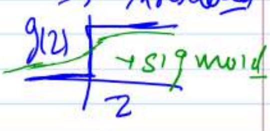
COL 774
Oct 15, 2025

Neural Networks:-

$x_0 = 1$, $x_1, \dots, x_m$ $\to$ $\sum_{j=0}^{n} \theta_j x_j \to \varphi \to h_\theta(x)$

$g(\theta^T x)$
↳ Activation function

$g \leftarrow \mathbb{1}\{\theta^T x \geq 0\}$
↳ Sigmoid $(\theta^T x)$
eg $\sigma(\theta^T x)$



$$J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} \left[y^{(i)} - h_\theta(x^{(i)}) \right]^2 \qquad \{x^{(i)}, y^{(i)}\}_{i=1}^{m}$$

$z = \theta^T x$

$$\nabla_\theta J(\theta) = \frac{2}{2m} (-1) \sum_{i=1}^{m} \left(y^{(i)} - h_\theta(x^{(i)}) \right) \nabla_\theta h_\theta(x^{(i)})$$

$\nabla_\theta g(\theta^T x^{(i)}) \Rightarrow \frac{\partial}{\partial z^{(i)}} g(z)^{(i)} \cdot \nabla_\theta (\theta^T x)^{(i)}$, $z^{(i)}$

$$\nabla_\theta J(\theta) = \frac{1}{2m} \sum_{i=1}^{m} \left(y^{(i)} - h_\theta(x^{(i)}) \right) h_\theta(x^{(i)})\left(1 - h_\theta(x^{(i)})\right) \cdot \nabla_\theta (\theta^T x^{(i)})$$

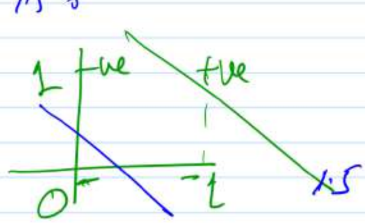
$$= \frac{1}{m} \sum_{i=1}^{m} (-1)\left(y^{(i)} - h_\theta(x^{(i)}) \right) \left(h_\theta(x^{(i)}) \right)\left(1 - h_\theta(x^{(i)})\right) x^{(i)}$$

Parameter learning algorithm:-

$t \leftarrow 0;$
$\theta^{(t)} \leftarrow \text{init}();$
do {
$\theta^{(t+1)} \leftarrow \theta^{(t)} - \eta \cdot \nabla_\theta J(\theta)$
$t \leftarrow t+1;$
} (while !converged)

⇒ Composing neurons/perceptrons:-

$x_1, x_2 \in \{0, 1\}$
$y \in \{0, 1\}$
$y = x_1 \wedge x_2$

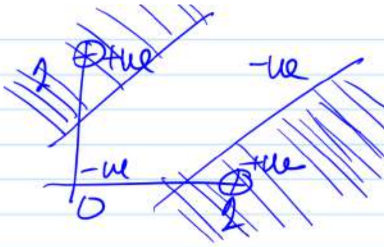


$h_\theta(x) = \mathbb{1}\{\theta_2 x_2 + \theta_1 x_1 + \theta_0 \geq 0\}$
⇓ ↳ 1 ↳ 1 ↳ −1.5
$\theta_2, \theta_1, \theta_0$

$\theta_2 x_2 + \theta_1 x_1 + \theta_0 = 0$

$y = x_1 \oplus x_2$?

$\Rightarrow (x_1 \wedge \neg x_2) \vee (\neg x_1 \wedge x_2)$



$z_1 = x_1 \wedge \neg x_2$

$x_1, x_2 \to \boxed{\ } \to \boxed{3} \to z_1 \vee z_2$

$z_2 = \neg x_1 \wedge x_2$ → output layer

→ hidden layer

$x_1 \dots x_n > \square \; \dots \; \square$ → ℓth layer in the network

→ kth neuron in ℓth layer

$n^{(\ell)}$:- # of neurons in layer ℓ.

layers $\ell \in \{1 \dots L\}$

$$\Theta^{(\ell)} = \begin{array}{c c} & n^{(\ell-1)} \\ n^{(\ell)} & \begin{bmatrix} & & \\ & \theta_{kj}^{(\ell)} & \\ & & \end{bmatrix} \end{array}$$

$o^{(\ell)}$:- output of neurons in the ℓth layer

output of kth neuron in ℓth layer $\leftarrow$ $o_k^{(\ell)} = g(\theta_k^{(\ell)T} o^{(\ell-1)})$ — ①

$o^{(\ell)} = g(\Theta^{(\ell)} o^{(\ell-1)})$ — ②

↳ point-wise

$net_k^{(\ell)} = \theta_k^{(\ell)T} o^{(\ell-1)}$

$o_k^{(\ell)} = g(net_k^{(\ell)})$

:- in the last layer.

$\hat{y}_1 \dots \hat{y}_r$ $\qquad \hat{y}_k \equiv o_k^{(L)}$

$\Theta = \{\Theta^{(\ell)}\}_{\ell=1}^{L}$ $\qquad \{x^{(i)}, y^{(i)}\}_{i=1}^{m}$

$$J(\Theta) = \frac{1}{2m} \sum_{i=1}^{m} \sum_{k=1}^{r} [y_k^{(i)} - \hat{y}_k^{(i)}]^2$$

$$\Theta^* = \arg\min_{\Theta} J(\Theta)$$

Assume $m = \underline{1}$